Algorithmic Features and Ablation Experiments

1. Impacts of Algorithmic Features

1.1 Uniform weights

The following figure, comparing the computation times of the five algorithmic configurations conducted on the 14 most challenging uniform-weight instances in S_A.

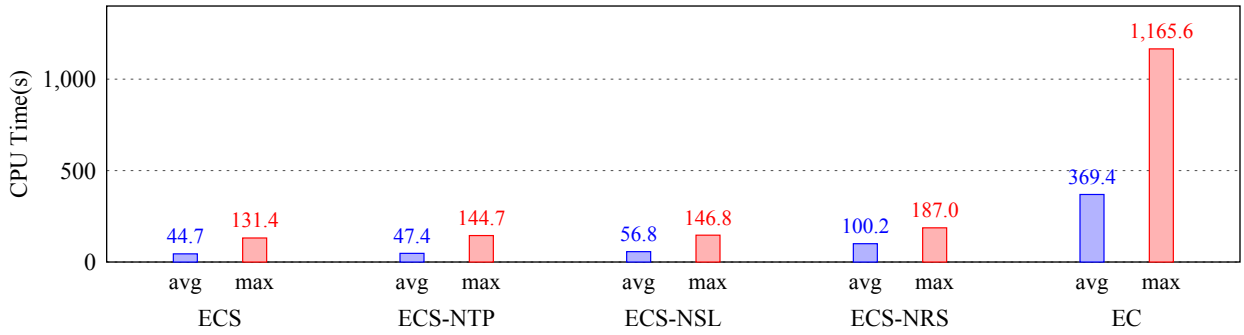


Figure 1 Computation Times of the Five Algorithm Configurations on the 14 Most Challenging Uniform-Weight Instances

The following table, comparing the number of tours enumerated for EC and ECS conducted on the 14 most challenging uniform-weight instances in S_A).

Table 1 Number of Tours Comparison between EC and ECS on 14 Uniform-Weight Instances

	EC						ECS					
Name	Cost	$\|\mathcal{U}^1\|$	$\|\mathcal{U}^2\|$	$\|\mathcal{U}^3\|$	$\|\mathcal{U}^4\|$	CPU	Cost	$\|\mathcal{U}^1\|$	$\|\mathcal{U}^2\|$	$\|\mathcal{U}^3\|$	$\|\mathcal{U}^4\|$	CPU
120-30-0.5-1	6695	8781	6018181	191071	143970	238.1	6695	5473	1090419	198302	139219	119.3
120-30-0.5-2	6747	13072	13361355	149785	104041	378.8	6747	7038	925997	60165	35868	45.9
120-30-0.5-3	6652	14059	6286077	24913	21870	197.5	6652	9314	871067	28071	22936	41.4
120-30-0.5-4	6679	18074	11214317	14332	10316	446.6	6679	8340	619692	23714	13766	27.1
120-30-0.6-1	6662	12901	9229395	225592	143967	310.4	6662	7242	1028148	107258	70282	52.1
120-30-0.6-2	6732	13267	11460121	25112	15126	257.7	6732	6590	182479	11689	11130	10.3
120-30-0.6-3	6635	16161	8410789	305221	161873	273.8	6635	10533	806259	82834	47287	42.3
120-30-0.6-4	6648	23045	12894824	34065	27379	575.0	6648	12584	680064	22271	17787	32.5
120-30-0.6-5	6578	14441	10848580	127442	103043	331.8	6578	7763	463945	88998	61995	32.6
120-30-0.7-1	6656	13622	8505618	309164	258747	330.2	6656	7407	900658	92716	76690	41.1
120-30-0.7-2	6726	13750	12270599	223344	121611	316.7	6726	4422	662981	103924	66566	28.1
120-30-0.7-3	6633	13179	0	0	0	4.0	6633	5995	0	0	0	3.2
120-30-0.7-4	6639	18594	14684908	1370042	779443	1165.5	6639	9827	1461558	327383	269120	131.3
120-30-0.7-5	6565	16115	11617300	86051	65488	345.0	6565	7441	433512	68584	35897	17.1

Algorithmic Features and Ablation Experiments

1. Impacts of Algorithmic Features

1.1 Uniform weights

Table 2 Detailed Results of ECS-NTP, ECS-NSL, and ECS-NRS on the 14 Most Challenging Uniform-Weight Instances in S_A

ECS-NTP											
ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
120-30-1	6695	0.00	137.64	120-30-1	6662	0.00	54.64	120-30-1	6656	0.00	41.03
120-30-2	6747	0.00	47.80	120-30-2	6732	0.00	10.39	120-30-2	6726	0.00	28.45
120-30-3	6652	0.00	40.84	120-30-3	6635	0.00	45.21	120-30-3	6633	0.00	3.30
120-30-4	6679	0.00	26.73	120-30-4	6648	0.00	32.17	120-30-4	6639	0.00	144.74
120-30-5	–	–	–	120-30-5	6578	0.00	32.04	120-30-5	6565	0.00	17.97
ECS-NSL											
ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
120-30-1	6695	0.00	107.68	120-30-1	6662	0.00	73.74	120-30-1	6656	0.00	53.51
120-30-2	6747	0.00	56.41	120-30-2	6732	0.00	29.21	120-30-2	6726	0.00	53.00
120-30-3	6652	0.00	46.20	120-30-3	6635	0.00	49.06	120-30-3	6633	0.00	3.15
120-30-4	6679	0.00	36.62	120-30-4	6648	0.00	69.99	120-30-4	6639	0.00	146.79
120-30-5	–	–	–	120-30-5	6578	0.00	31.80	120-30-5	6565	0.00	37.42
ECS-NRS											
ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
120-30-1	6695	0.00	150.19	120-30-1	6662	0.00	97.32	120-30-1	6656	0.00	92.36
120-30-2	6747	0.00	108.13	120-30-2	6732	0.00	94.86	120-30-2	6726	0.00	108.80
120-30-3	6652	0.00	101.40	120-30-3	6635	0.00	101.20	120-30-3	6633	0.00	3.11
120-30-4	6679	0.00	124.03	120-30-4	6648	0.00	126.42	120-30-4	6639	0.00	186.99
120-30-5	–	–	–	120-30-5	6578	0.00	60.02	120-30-5	6565	0.00	47.98

Algorithmic Features and Ablation Experiments

1. Impacts of Algorithmic Features

1.2 Standard weights

The following figure, comparing the computation times of the five algorithmic configurations conducted on the 14 most challenging standard-weight instances in S_A.

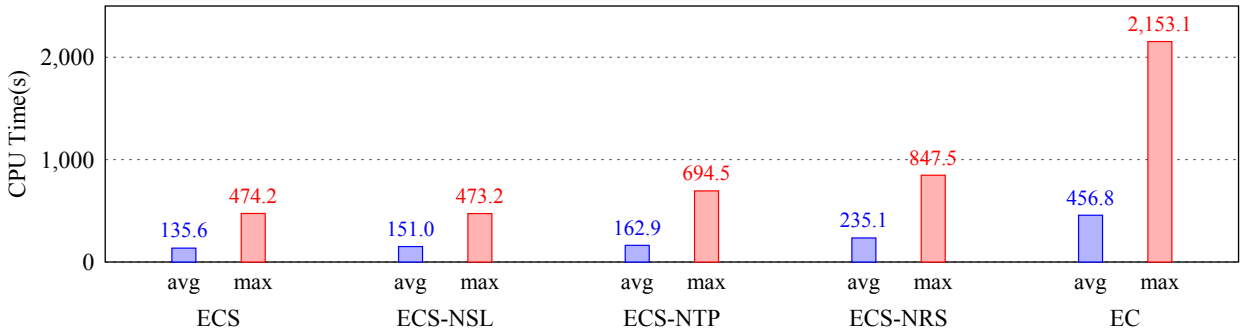


Figure 2 Computation Times of the Five Algorithm Configurations on the 14 Most Challenging Standard-Weight Instances

The following table, comparing the number of tours enumerated for EC and ECS conducted on the 14 most challenging standard-weight instances in S_A).

Table 3 The Number of Tours Comparison between EC and ECS on the 14 Most Challenging Standard-Weight Instances

	EC						ECS					
Name	Cost	$\|\mathcal{U}^1\|$	$\|\mathcal{U}^2\|$	$\|\mathcal{U}^3\|$	$\|\mathcal{U}^4\|$	CPU	Cost	$\|\mathcal{U}^1\|$	$\|\mathcal{U}^2\|$	$\|\mathcal{U}^3\|$	$\|\mathcal{U}^4\|$	CPU
120-30-0.5-1	13038	8077	5979998	148462	109488	229.5	13038	5399	1008839	167166	121729	99.2
120-30-0.5-2	11922	12328	12766186	16082	12767	316.3	11922	12328	12766186	16082	12767	326.7
120-30-0.5-3	11191	13666	6044254	34521	27747	161.8	11191	10234	856061	29653	21488	47.0
120-30-0.5-4	11440	19859	11893932	11081	8564	456.6	11440	19859	11893932	11081	8564	474.1
120-30-0.6-1	12967	11552	9077630	73302	58522	305.4	12967	7027	1039136	92586	64516	57.0
120-30-0.6-2	11896	13343	11473223	16327	12379	273.4	11896	6105	146727	13473	8605	9.3
120-30-0.6-3	11156	14967	8190603	368086	176319	271.2	11156	10227	813807	95361	79742	45.0
120-30-0.6-4	11382	26728	12749952	56593	27534	572.5	11382	13199	929176	73184	25016	47.2
120-30-0.6-5	13314	16217	10297040	46370	34087	304.0	13314	7132	423032	62877	37942	29.3
120-30-0.7-1	12955	12324	8491512	195709	67217	306.4	12955	6929	719502	61302	61293	30.4
120-30-0.7-2	11886	11959	12010382	213092	167224	405.3	11886	11959	12010382	213092	167224	424.1
120-30-0.7-3	11152	11197	8063777	398449	212516	257.5	11152	7201	0	0	0	3.2
120-30-0.7-4	11366	26728	14887523	2472178	M	2153.1(M)	11366	10356	1511057	714172	604041	271.5
120-30-0.7-5	13285	14851	11693291	220091	160989	382.2	13285	6747	516305	147966	90602	33.2

Algorithmic Features and Ablation Experiments

1. Impacts of Algorithmic Features

1.2 Standard weights

Table 4 Detailed Results of ECS-NTP, ECS-NSL, and ECS-NRS on the 14 Most Challenging Standard-Weight Instances in S_A

ECS-NTP											
ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
120-30-1	13038	0.00	133.86	120-30-1	12967	0.00	68.28	120-30-1	12955	0.00	40.15
120-30-2	11922	0.00	334.31	120-30-2	11896	0.00	10.42	120-30-2	11886	0.00	694.49
120-30-3	11191	0.00	57.77	120-30-3	11156	0.00	51.10	120-30-3	11152	0.00	5.61
120-30-4	11440	0.00	475.59	120-30-4	11382	0.00	52.43	120-30-4	11366	0.00	286.93
120-30-5	–	–	–	120-30-5	13314	0.00	33.14	120-30-5	13285	0.00	37.02
ECS-NSL											
ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
120-30-1	13038	0.00	101.27	120-30-1	12967	0.00	88.34	120-30-1	12955	0.00	71.22
120-30-2	11922	0.00	332.13	120-30-2	11896	0.00	38.51	120-30-2	11886	0.00	473.24
120-30-3	11191	0.00	44.98	120-30-3	11156	0.00	49.93	120-30-3	11152	0.00	3.29
120-30-4	11440	0.00	466.38	120-30-4	11382	0.00	78.83	120-30-4	11366	0.00	284.21
120-30-5	–	–	–	120-30-5	13314	0.00	31.32	120-30-5	13285	0.00	50.88
ECS-NRS											
ws-0.5				ws-0.6				ws-0.7			
Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU	Name	Cost	Gap‰	CPU
120-30-1	13038	0.00	159.42	120-30-1	12967	0.00	101.84	120-30-1	12955	0.00	85.64
120-30-2	11922	0.00	558.20	120-30-2	11896	0.00	98.40	120-30-2	11886	0.00	847.53
120-30-3	11191	0.00	114.75	120-30-3	11156	0.00	115.07	120-30-3	11152	0.00	3.42
120-30-4	11440	0.00	616.30	120-30-4	11382	0.00	147.95	120-30-4	11366	0.00	319.65
120-30-5	–	–	–	120-30-5	13314	0.00	61.45	120-30-5	13285	0.00	61.20

Algorithmic Features and Ablation Experiments

2. Observations on ECS

2.1 Uniform weights

The following table, containing the summarized results for the uniform-weight instances in S_A:

Table 5 Summary of the Results for Uniform-Weight Instances in S_A

			Step 1			*Step* 2	*Step* 3			*Step* 4		
Name	Inst	Switch	LB‰	$\|\mathcal{T}^1\|$	$\|\mathcal{U}^1\|$	UB‰	$\|\mathcal{U}^2\|$	$\|\mathcal{U}^3\|$	$\|\mathcal{U}^4\|$	$\|\hat{\mathcal{T}}\|$	$\|\hat{\mathcal{U}}\|$	CPU
60-10-0.5	5	0	4.8	5.8	130.6	0.0	85.2	39.4	29.0	0.0	0.0	0.1
60-10-0.6	5	0	3.1	5.2	131.0	0.0	63.6	31.4	25.4	0.0	0.0	0.1
60-10-0.7	5	0	6.2	4.0	100.0	0.0	91.6	54.6	40.6	0.0	0.0	0.1
60-20-0.5	5	0	2.1	8.0	328.8	0.0	250.0	81.6	60.0	0.0	0.0	0.2
60-20-0.6	5	1	2.3	7.6	688.4	0.0	1053.8	142.0	73.6	0.0	0.0	0.3
60-20-0.7	5	0	0.4	7.0	400.0	0.0	92.4	39.2	36.2	0.0	0.0	0.1
60-30-0.5	2	0	4.1	12.5	683.5	0.0	2070.0	517.0	422.5	0.0	0.0	0.6
60-30-0.6	5	0	3.0	12.0	832.2	0.0	2172.4	557.4	460.8	0.0	0.0	0.7
60-30-0.7	5	0	2.0	10.6	881.6	0.0	1853.6	486.0	407.2	0.0	0.0	0.5
90-10-0.5	4	0	6.2	8.0	270.3	0.0	279.3	118.8	98.3	0.0	0.0	0.1
90-10-0.6	5	0	4.0	6.6	259.6	0.0	219.4	105.2	72.4	0.0	0.0	0.1
90-10-0.7	5	0	3.8	5.4	258.2	0.0	315.6	139.4	98.8	0.0	0.0	0.1
90-20-0.5	4	0	2.9	13.0	1063.3	0.0	3868.8	511.0	401.3	0.0	0.0	0.6
90-20-0.6	5	0	2.1	11.8	999.2	0.0	3677.6	714.6	592.8	0.0	0.0	0.5
90-20-0.7	5	1	0.9	9.8	1750.0	0.6	18396.0	1413.4	994.4	0.4	7.2	0.7
90-30-0.5	5	2	4.8	17.4	4984.2	0.0	325016.0	15450.4	11397.2	0.0	0.0	17.4
90-30-0.6	5	1	2.4	15.4	3918.4	0.0	144270.6	9415.4	7499.6	0.0	0.0	5.2
90-30-0.7	5	0	0.9	14.4	2756.6	0.0	17331.6	3083.6	2823.2	0.0	0.0	1.7
120-10-0.5	4	1	12.8	12.0	1268.0	0.0	14553.5	4681.8	3850.8	0.0	0.0	1.6
120-10-0.6	5	0	7.2	9.0	646.8	0.0	5952.6	2503.4	2109.6	0.0	0.0	0.5
120-10-0.7	5	0	5.5	7.2	642.0	0.0	4274.2	2376.8	1494.2	0.0	0.0	0.3
120-20-0.5	4	1	9.6	17.8	3788.3	1.1	301130.0	26839.8	20790.8	0.0	0.0	16.9
120-20-0.6	5	0	2.5	14.4	2825.2	0.5	54099.0	13221.2	10006.0	0.0	0.0	3.6
120-20-0.7	5	0	3.0	12.2	2349.8	0.0	68545.2	16788.8	14045.0	0.0	0.0	3.6
120-30-0.5	4	0	3.7	21.3	7541.3	0.0	876793.8	77563.0	52947.3	0.0	0.0	58.5
120-30-0.6	5	0	1.6	19.6	8942.4	0.0	632179.0	62610.0	41696.2	0.0	0.0	34.0
120-30-0.7	5	0	1.6	18.6	7018.4	0.0	691741.8	118521.4	89654.6	0.0	0.0	44.2

The following table, containing the summarized results for the standard-weight instances in S_A:

Table 6 Summary of the Results for Standard-Weight Instances in S_A

			Step 1			*Step* 2	*Step* 3			*Step* 4		
Name	Inst	Switch	LB‰	$\lvert\mathcal{T}^1\rvert$	$\lvert\mathcal{U}^1\rvert$	UB‰	$\lvert\mathcal{U}^2\rvert$	$\lvert\mathcal{U}^3\rvert$	$\lvert\mathcal{U}^4\rvert$	$\lvert\hat{\mathcal{T}}\rvert$	$\lvert\hat{\mathcal{U}}\rvert$	CPU
60-10-0.5	5	0	5.0	6.2	132.8	0.0	81.6	38.0	31.6	0.0	0.0	0.1
60-10-0.6	5	0	4.5	5.2	121.0	0.0	77.2	38.2	35.8	0.0	0.0	0.1
60-10-0.7	5	0	3.6	4.6	101.0	0.0	81.2	40.6	33.2	0.0	0.0	0.1
60-20-0.5	5	0	3.0	8.4	447.2	0.0	159.4	52.0	48.0	0.0	0.0	0.2
60-20-0.6	5	1	1.1	7.6	750.2	0.0	110.0	37.6	25.4	0.0	0.0	0.2
60-20-0.7	5	0	0.3	7.0	421.2	0.0	39.2	15.0	9.6	0.0	0.0	0.1
60-30-0.5	2	0	4.6	12.0	746.5	0.0	1933.5	444.0	339.0	0.0	0.0	0.6
60-30-0.6	5	0	1.8	10.4	847.0	1.4	1559.6	525.6	451.0	0.0	0.0	0.7
60-30-0.7	5	1	1.5	9.6	1465.2	0.0	933.6	271.4	232.4	0.0	0.0	0.6
90-10-0.5	4	0	9.5	7.8	278.8	0.0	281.8	154.8	136.3	0.0	0.0	0.1
90-10-0.6	5	0	3.9	6.8	265.8	0.0	207.4	89.6	68.2	0.0	0.0	0.1
90-10-0.7	5	0	5.6	6.0	260.8	0.7	367.6	210.8	184.4	0.0	0.0	0.1
90-20-0.5	4	0	2.3	13.3	1074.5	0.0	3689.0	433.5	342.5	0.0	0.0	0.7
90-20-0.6	5	0	2.3	12.4	1073.0	0.0	2961.4	900.0	766.6	0.0	0.0	0.6
90-20-0.7	5	1	0.3	11.0	1905.4	0.3	17278.6	1468.2	1162.4	0.0	0.0	0.8
90-30-0.5	5	1	3.4	17.2	3695.4	0.6	213176.4	8205.2	6130.0	0.0	0.0	10.2
90-30-0.6	5	2	1.7	15.6	5053.8	0.2	201542.4	4919.2	3727.8	0.0	0.0	6.2
90-30-0.7	5	0	1.6	14.2	2588.8	0.0	19322.8	4950.4	4555.4	0.0	0.0	1.9
120-10-0.5	4	1	13.5	11.8	1161.8	1.3	12578.5	3710.0	2881.5	0.0	0.0	2.2
120-10-0.6	5	0	9.1	9.6	665.0	0.0	5233.8	2510.4	2241.8	0.0	0.0	0.5
120-10-0.7	5	0	7.9	7.6	688.0	0.0	4084.8	2812.4	2403.4	0.0	0.0	0.4
120-20-0.5	4	1	8.4	18.0	3811.3	0.5	270699.3	11684.5	9843.5	0.0	0.0	12.1
120-20-0.6	5	0	2.5	14.2	2904.0	0.0	44484.2	12432.8	10639.8	0.0	0.0	3.7
120-20-0.7	5	0	4.1	12.0	2338.2	0.0	74247.4	26794.6	21685.6	0.0	0.0	5.8
120-30-0.5	4	2	3.1	21.3	11955.0	0.2	6631254.5	55995.5	41137.0	0.0	0.0	236.8
120-30-0.6	5	0	1.7	18.2	8738.0	0.0	670375.6	67496.2	43164.2	0.0	0.0	37.6
120-30-0.7	5	1	1.8	16.6	8638.4	0.2	2951449.2	227306.4	184632.0	0.4	193.8	152.5